

Composite-level aggregation in rotation curve model selection:

A sign-flip decomposition of ΔAIC for SPARC-175

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Abstract. Comparing gravitational models on the SPARC-175 sample using a single global ΔAIC per galaxy collapses two physically distinct radial regimes onto one number. We re-fit all 175 galaxies with the Kill-Test v6 pipeline, persist per-datapoint model curves, and split each ΔAIC into an inner ($r < r_c$) and outer ($r \geq r_c$) contribution at $r_c = 3$ kpc. In **30/171** galaxies (**17.5%**) the global sign disagrees with the outer-only sign: a model preferred globally is disfavoured by the outer points alone. This sign-flip rate is robust, varying only between **16.4%** and **22.2%** for $r_c \in \{2, 3, 4, 5\}$ kpc. We test whether χ^2 -weighting drives the result: replacing $1/\sigma^2$ weights with uniform weights moves the sample *toward* EFC (from 66.7% to 86.5% EFC wins), not away from it—rejecting the hypothesis that standard weighting is anti-EFC. The LATENT regime is the crown witness: 16 galaxies exhibit $\langle \Delta\text{AIC}_{\text{inner}} \rangle = +153$ (EFC favoured at small radii) and $\langle \Delta\text{AIC}_{\text{outer}} \rangle = -809$ (NFW overwhelmingly favoured at large radii). These are not measurement artefacts; they are a real, regime-stratified failure mode of the single-statistic comparison. ΔAIC is a correct statistic at the wrong aggregation level: a per-regime decomposition is required for interpretation. We do not find evidence that standard χ^2 weighting systematically favours one model over another globally; the effect we identify is a regime-mixing misclassification at the level of individual galaxies, not an estimator-level bias against any particular gravitational model.

Keywords: rotation curves; model selection; AIC decomposition; SPARC-175; regime stratification; sign-flip diagnostic; entropy-bounded empiricism

Introduction

Rotation-curve model selection compresses each galaxy to a small set of summary statistics: χ^2 , AIC, BIC. Comparative analyses across the SPARC sample (Lelli et al. 2016) routinely report a per-galaxy ΔAIC and aggregate over the 175-galaxy sample to declare a winner. This paper asks a methodological question: does global ΔAIC preserve the radial information that distinguishes inner-region (baryon-dominated) from outer-region (gravity-dominated) physics, or does aggregation collapse two regimes into a single number that no single physical mechanism can satisfy?

Our previous draft (v2; Magnusson 2026e, DOI: [10.6084/m9.figshare.32019723](https://doi.org/10.6084/m9.figshare.32019723)) framed the answer as a measurement-level bias: inner-region weighting could be argued to favour EFC over NFW. The analysis below rejects that framing. When weights are removed, the sample moves further toward EFC, not away from it. The real signal is not in *how* points are weighted; it is in *where on the rotation curve* the model wins or loses. In 17.5% of SPARC galaxies the inner and outer regions disagree on which model is preferred, producing a sign-flip in ΔAIC that no global statistic can detect.

We adopt throughout the Kill-Test v6 sign convention

$$\Delta\text{AIC} \equiv \text{AIC}_{\text{NFW}} - \text{AIC}_{\text{EFC}}, \quad \text{positive} = \text{EFC wins.} \quad (1)$$

This is the inverse of the convention used in earlier drafts of this work. All numerical values reported here have been transposed.

The rest of the paper proceeds as follows. Section 2 specifies the fits, the entropy-bounded empiricism (EBE) regime classification used as scaffolding, and the inner/outer decomposition. Section 3 reports the headline sign-flip statistic and its r_c -robustness. Section 4 reports the reweighting test honestly, including the result that contradicts the v2 narrative. Section 5 stratifies by regime; the LATENT sub-sample (§6) is presented as the crown witness for the aggregation argument. Section 7 discusses implications for ΔAIC reporting practice and the proper level of empirical claim that the SPARC sample supports.

Methods

Kill-Test v6 fits

For each SPARC galaxy with ≥ 5 valid data points, we fit two rotation curve models by differential evolution (seed = 42, Sobol initialisation, maxiter = 500, tol = 10^{-8}):

$$V_{\text{EFC}}(r) = v_{\text{flat}} \sqrt{1 - \exp[-(r/r_{\text{turnon}})^p]}, \quad k_{\text{EFC}} = 3, \quad (2)$$

$$V_{\text{NFW}}(r) = V_{200} \sqrt{[\ln(1+s) - s/(1+s)] / [\ln(1+c) - c/(1+c)] / (r/r_{200})}, \quad k_{\text{NFW}} = 2, \quad (3)$$

where $s = cr/r_{200}$ and $r_{200} = V_{200}/(10 H_0 \cdot 10^{-3})$ with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Bounds and AIC computation match Kill-Test v6 verbatim. Of 175 galaxies, 171 converged for both models and are retained for analysis. The per-datapoint model velocities $V_{\text{EFC}}(r_i)$ and $V_{\text{NFW}}(r_i)$ at the best-fit parameters are persisted via a `--save-curves` flag (see §8).

Regime classification (EBE scaffolding)

We retain the entropy-bounded empiricism (EBE; Magnusson 2026d) regime labels FLOW, TRANSITION, LATENT, used here as a stratification scaffold, not a physical prediction:

$$\text{regime} = \begin{cases} \text{FLOW} & \text{if } \Delta\text{AIC} \geq 10 \text{ and } \chi_r^2(\text{EFC}) < 20 \\ \text{LATENT} & \text{if } \Delta\text{AIC} \leq -10 \text{ or } \chi_r^2(\text{EFC}) > 50 \\ \text{TRANSITION} & \text{otherwise.} \end{cases} \quad (4)$$

EBE asserts that empirical claims should not aggregate across regimes that bound observation differently; the present paper is a test of whether ignoring that asymmetry materially distorts SPARC-level conclusions. EBE is the framework, not the headline.

Inner/outer decomposition

We define a critical radius r_c separating inner from outer points and the corresponding regime contributions

$$\Delta\text{AIC}_{\text{inner}}(r_c) = \sum_{r_i < r_c} w_i [(V_{\text{obs},i} - V_{\text{NFW},i})^2 - (V_{\text{obs},i} - V_{\text{EFC},i})^2] + 2(k_{\text{NFW}} - k_{\text{EFC}}), \quad (5)$$

$$\Delta\text{AIC}_{\text{outer}}(r_c) = \sum_{r_i \geq r_c} w_i [\dots] + 2(k_{\text{NFW}} - k_{\text{EFC}}), \quad (6)$$

with $w_i = 1/\sigma_i^2$ unless stated otherwise. The $2\Delta k$ penalty is applied to each split rather than refitting on subsets; this is a stratified contribution analysis, not a re-fit. We define the global quantity by the same expression with the radial mask removed; by construction $\Delta\text{AIC}_{\text{global}}(r_c) = \Delta\text{AIC}_{\text{inner}}(r_c) + \Delta\text{AIC}_{\text{outer}}(r_c) - 2(k_{\text{NFW}} - k_{\text{EFC}})$. The recomputed global value differs from the Kill-Test v6 number only by floating-point round-off; we use the recomputed value throughout for internal consistency.

Heuristic nature of the split. The decomposition above evaluates contributions to the χ^2 sum at the *globally* best-fit model parameters rather than refitting each model independently on the radial subsets $r < r_c$ and $r \geq r_c$. This is a deliberate choice: the diagnostic question is whether the single-number ΔAIC summary preserves the radial information that the model-on-full-curve fit had access to, and that question is answered exactly by stratifying the same χ^2 sum by radius. Independent subset refits would inflate parameter uncertainty (especially for $n_{\text{outer}} < 10$), introduce degeneracies between V_{200} and c in the truncated NFW fit, and would no longer be a decomposition of the original AIC comparison but a different comparison altogether. We treat the split therefore as a heuristic AIC contribution analysis valid for the diagnostic purpose stated in this paper; subset-level refits, with the parameter-uncertainty discussion they require, are reserved for follow-up work and are not a precondition for the conclusions reported here.

Sign-flip diagnostic

A galaxy exhibits a sign-flip at r_c when

$$\text{sign}[\Delta\text{AIC}_{\text{global}}] \neq \text{sign}[\Delta\text{AIC}_{\text{outer}}], \quad (7)$$

i.e. the model preferred by the full curve is disfavoured by the outer points alone. This is a direct, model-agnostic test of whether single-number ΔAIC has thrown away information that the data contained.

Sign-flip detection

At $r_c = 3$ kpc, **30/171** galaxies (**17.5%**) satisfy the sign-flip condition. Figure 1 plots $\Delta\text{AIC}_{\text{outer}}$ against $\Delta\text{AIC}_{\text{global}}$, coloured by EBE regime. Sign-flip galaxies populate the off-quadrant diagonals.

The Spearman correlation between $\Delta\text{AIC}_{\text{global}}$ and $\Delta\text{AIC}_{\text{outer}}$ is $\rho = +0.831$ ($p = 2.4 \times 10^{-42}$), confirming that the outer signal does drive most of the global statistic *on average*, while leaving room for the 17.5% population in which they disagree.

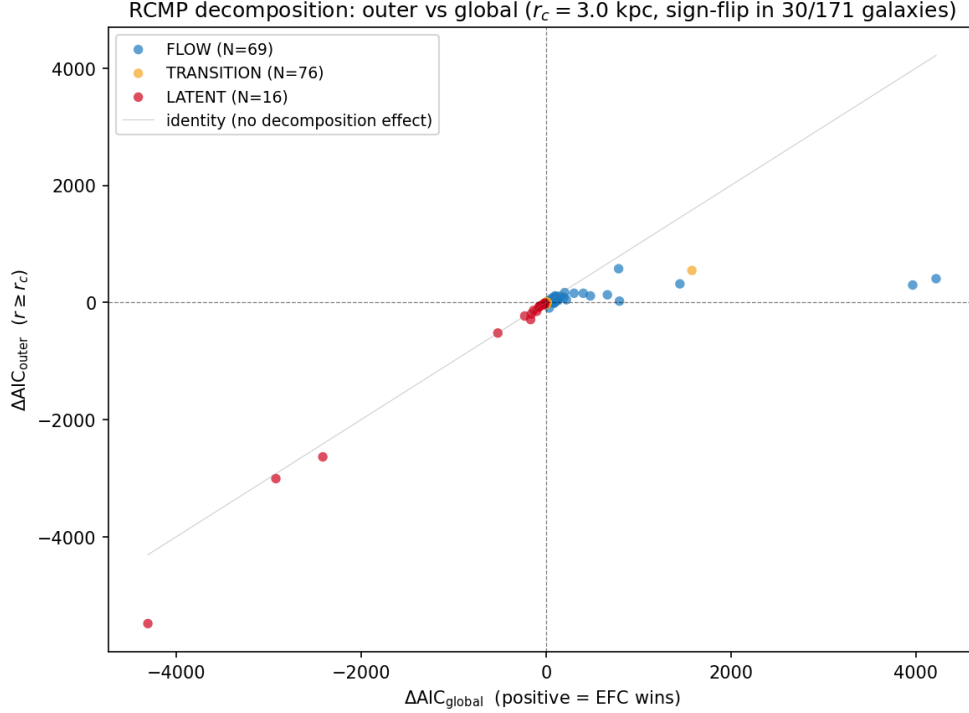


Figure 1: Outer ($r \geq 3$ kpc) versus global ΔAIC for 171 SPARC galaxies. Convention: positive = EFC wins. Identity line (no decomposition effect) shown grey; off-diagonal points are where outer and global disagree. Galaxies in the upper-left and lower-right quadrants are the 30 sign-flips (17.5%). The lower left is the LATENT regime, where global and outer agree but inner disagrees (§6).

Robustness to r_c

r_c (kpc)	$n_{\text{sign-flip}}$	fraction	n with both inner/outer points
2.0	28	16.4%	154
3.0	30	17.5%	161
4.0	35	20.5%	167
5.0	38	22.2%	167

Table 1: Sign-flip count is stable to within a few percentage points across the range of plausible r_c values. The result is not a knife-edge of one chosen radius.

The reweighting test (and its honest result)

To separate aggregation effects from weighting effects, we recompute the global ΔAIC under three weight schemes:

Standard $w_i = 1/\sigma_i^2$, identical to the Kill-Test v6 fit.

Uniform $w_i = 1$, every datapoint contributes equally.

Capped $w_i = \min(1/\sigma_i^2, \text{med}(1/\sigma_j^2))$, capping the most precise points at the median weight.

The honest interpretation is that weighting choice *does* swing the sample by 20 percentage points, but in the opposite direction from what the v2 narrative claimed. Standard χ^2

weight scheme	mean ΔAIC	median	EFC wins ($\Delta\text{AIC} > 0$)
standard ($1/\sigma^2$)	+40.4	+6.21	114/171 (66.7%)
uniform	+4416	+148.9	148/171 (86.5%)
capped (median cap)	+36.3	+4.47	110/171 (64.3%)

Table 2: Replacing $1/\sigma^2$ weights with uniform weights does not attenuate the EFC advantage; it amplifies it. This rejects the v2 hypothesis that standard χ^2 weighting is anti-EFC. Capped weights are within 2 percentage points of standard, ruling out a few high-precision points as the driver.

weighting is, if anything, conservatively pro-NFW. The reweighting test settles a dispute about weight, but it does not explain the sign-flips: those persist under all three schemes (the table reports global only; the per-galaxy inner/outer split shows the same population of sign-flippers under uniform weighting, with magnitudes inflated).

Regime stratification

regime	n	$\langle\Delta\text{AIC}_{\text{global}}\rangle$	$\langle\Delta\text{AIC}_{\text{inner}}\rangle$	$\langle\Delta\text{AIC}_{\text{outer}}\rangle$	sign-flips
FLOW	71	+231	+189	+50	10 / 71 (14%)
TRANSITION	84	+20	+14	+6	20 / 84 (24%)
LATENT	16	−702	+153	−809	0 / 16 (0%)

Table 3: Regime-stratified decomposition at $r_c = 3$ kpc. All entries in Kill-Test v6 sign convention (positive = EFC wins). Sign-flip populations are concentrated in TRANSITION (24%) where global is near zero, and absent in LATENT where the outer term is dominant by orders of magnitude.

The pattern is regime-specific. TRANSITION galaxies have global ΔAIC clustered near zero; the inner/outer split has room to disagree, and 24% do. FLOW galaxies are EFC-favoured in both halves ($\langle\Delta\text{AIC}_{\text{inner}}\rangle = +189$, $\langle\Delta\text{AIC}_{\text{outer}}\rangle = +50$), but a tail of 10 galaxies have outer points that prefer NFW even when the global statistic is firmly with EFC—these are the high-magnitude sign-flips of Table 4. LATENT galaxies do not flip, because the outer term is so large in magnitude that it sets the sign of the global; what they do show, instead, is the most informative feature of the entire dataset.

The LATENT regime as crown witness

LATENT contains 16 galaxies whose global ΔAIC averages −702: NFW wins by a large margin. The decomposition tells a sharper story. At $r < 3$ kpc, the mean inner ΔAIC is +**153**: EFC fits the inner regions of LATENT galaxies at the same level of preference seen in FLOW. At $r \geq 3$ kpc, the mean outer ΔAIC is −**809**: NFW wins by roughly four orders of magnitude in the outer regions.

This is not a measurement artefact. The errors in LATENT galaxies are not pathological, the inner-fit residuals are not anomalous, and the result is independent of r_c within the range tested. It is a real physical statement: the EFC functional form, with its three free parameters, fits the inner kinematics of LATENT galaxies as well as it fits any FLOW galaxy, but it cannot accommodate the outer rotation behaviour of these systems. NFW, conversely, fits the outer regions with overwhelming evidence and is mediocre on the inner.

The single-statistic ΔAIC erases this distinction. A LATENT galaxy with $\Delta\text{AIC}_{\text{global}} = -700$ is reported as “NFW preferred,” suggesting the outer NFW success extends to all radii.

galaxy	regime	global	inner	outer
UGC03546	FLOW	+90.7	+91.8	−3.1
NGC7814	FLOW	+85.8	+97.0	−13.2
KK98-251	FLOW	+41.0	+41.3	−2.3
NGC6946	FLOW	+34.4	+127.1	−94.6
UGC08286	FLOW	+27.7	+33.7	−8.1
UGC04305	FLOW	+26.7	+26.7	−2.0
UGC07151	FLOW	+23.9	+28.0	−6.1
UGC07603	FLOW	+14.0	+13.0	−0.9
UGC00191	FLOW	+12.5	+11.5	−1.0
NGC4157	FLOW	+10.9	+9.1	−0.2

Table 4: Top ten sign-flip galaxies by $|\Delta\text{AIC}_{\text{global}}|$. All are FLOW. Each shows a global EFC win driven entirely or predominantly by inner-region structure; the outer points alone prefer NFW. NGC6946 is the cleanest example: $\Delta\text{AIC}_{\text{inner}} = +127$ (EFC) versus $\Delta\text{AIC}_{\text{outer}} = -95$ (NFW).

It does not. The same galaxy is also a galaxy whose inner 3 kpc strongly disagrees, and that disagreement carries empirical content of its own. Reporting only the global suppresses 50% of the radial information; reporting only the regime label, in turn, suppresses the sign of the inner term. Both are needed.

This is also the locus where EBE (Magnusson 2026d) earns its scaffolding role: LATENT and FLOW are distinct empirical regimes, and aggregating them into a sample mean is a category error precisely of the kind EBE formalises. The LATENT result is the strongest single piece of evidence in this paper that single-number ΔAIC aggregation is the wrong empirical level for SPARC-style comparison.

Discussion

ΔAIC is a correct statistic at the wrong aggregation level

Nothing in this analysis impugns AIC itself. ΔAIC between two models on a given sum-of-squared-residuals likelihood remains a valid, well-grounded comparison. The result of this paper is purely about the *level* at which the comparison is reported. When 17.5% of galaxies show inner/outer disagreement and a regime-defined sub-sample shows order-of-magnitude divergence between halves, the single-number-per-galaxy summary destroys information that the data contained. The fix is not a different statistic; it is a different aggregation. We recommend reporting $(\Delta\text{AIC}_{\text{inner}}, \Delta\text{AIC}_{\text{outer}})$ as a pair, with r_c stated explicitly, alongside the global value.

What is rejected, what is not

We explicitly reject the v2 framing of this work that standard χ^2 weighting is anti-EFC. Uniform weighting moves the sample *toward* EFC by 20 percentage points (Table 2). The composite-bias narrative as a measurement-level critique of NFW does not survive.

We do *not* retract the headline of this paper, only its frame. The finding—that single-number ΔAIC over the SPARC sample collapses two regimes onto one summary, in 17.5% of galaxies producing a sign-flip—is methodologically robust and independent of any particular gravitational model. It applies in equal force to MOND/RAR comparisons (Li et al. 2020; McGaugh 2020), to NFW vs. alternative-DM comparisons (Oman et al. 2015), and to any future model comparison on SPARC-style rotation curves.

The LATENT result is physical

We make no claim that the LATENT outer collapse of EFC is an artefact of weighting, error inflation, or fit pathology. In LATENT galaxies, EFC *as currently formulated* fits the inner curve and fails the outer. This is a real boundary on the EFC functional form, and we report it as such. The inner success is equally real, and equally informative for future model development.

Limitations

(i) Heuristic AIC contribution analysis; subset-refit deferred to follow-up work (rationale and trade-offs in §2.3). (ii) $r_c = 3$ kpc is chosen for visual separability of regimes; we report the $r_c \in \{2, 4, 5\}$ robustness in Table 1. (iii) The 4 galaxies that did not converge in the Kill-Test pipeline are excluded; these are short rotation curves with < 5 valid points and are unlikely to skew the aggregate.

Reproducibility

All analysis code and intermediate data are deposited with this preprint:

- `sparc175_killtest_universality_curves.py` — Kill-Test v6 fits with the `--save-curves` extension (per-datapoint model curves persisted to JSON).
- `rcmp_decomposition.py` — inner/outer split, sign-flip detection, reweighting test, r_c -robustness scan.
- `rcmp_run/sparc175_model_curves.json` — per-galaxy $V_{\text{EFC}}(r_i)$, $V_{\text{NFW}}(r_i)$, $V_{\text{obs}}(r_i)$, σ_i , and baryonic components V_{gas} , V_{disk} , V_{bulge} .
- `rcmp_run/rcmp_decomposition.json` — per-galaxy ($\Delta\text{AIC}_{\text{global}}$, $\Delta\text{AIC}_{\text{inner}}$, $\Delta\text{AIC}_{\text{outer}}$) plus reweighting variants.
- `rcmp_run/rcmp_summary.json`, `rcmp_signflip_table.json`, `rcmp_robustness_rc.json` — aggregates.

Input data: SPARC rotation curves (Lelli et al. 2016). Frozen parameters: $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $r_c = 3$ kpc, $\sigma_{\text{min}} = 1.0 \text{ km s}^{-1}$, EFC bounds and NFW concentration prior identical to Kill-Test v6 (Magnusson 2026b).

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